

Part I

SUMMARY of (In)determinate Forms

Summary

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Some notes on notation for finding limits:

- (a) **The first step in evaluating any limit is to find the form.** This is true, even if the question does not specifically ask for the form.
- (b) To find the form of $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$, take the limits of the numerator and denominator SEPARATELY. Do not just “plug in $x = a$ ”. “Plugging in” can not occur until after the function is noticed as continuous.
- (c) When writing a form of a limit, we NEVER write $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$. It is the $=$ that makes this a mathematically incorrect statement. Instead, write $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$.
- (d) When evaluating limits, do NOT drop the $\lim_{x \rightarrow a}$ until you have evaluated the limit. For example, the following is INCORRECT:

$$\lim_{x \rightarrow 2} \frac{2x^2 - 4x}{x - 2} = \frac{2x(x - 2)}{x - 2} = 2x = 4$$

Instead, we write:

$$\lim_{x \rightarrow 2} \frac{2x^2 - 4x}{x - 2} = \lim_{x \rightarrow 2} \frac{2x(x - 2)}{x - 2} = \lim_{x \rightarrow 2} 2x = 4$$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 Which of the following limits are of the form $\frac{0}{0}$?

(a) $\lim_{x \rightarrow 7} \frac{\sin(x-7)}{|x-7|}$

(b) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(x)}{x}$

(c) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{|\cos(x)|}{x}$

(d) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\pi - 2x}{\cos(x)}$

(e) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 4}$

(f) $\lim_{x \rightarrow 4} \frac{x^2 - 4}{x - 4}$

(g) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

(h) $\lim_{x \rightarrow 3} \frac{x - 4}{\sqrt{25 - x^2} - 4}$

(i) $\lim_{x \rightarrow 3} \frac{x - 3}{\sqrt{25 - x^2} - 4}$

Explanation. (a), (d), (g) and (i)

Problem 2

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Problem 2 State the form of the limit and then evaluate the limit.

$$(a) \lim_{x \rightarrow 6} \frac{4x^2 - 144}{x - 6}$$

Explanation. $\lim_{x \rightarrow 6} \frac{4x^2 - 144}{x - 6} = \lim_{x \rightarrow 6} \frac{4(x^2 - 36)}{x - 6}.$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 6} \frac{4x^2 - 144}{x - 6} &= \lim_{x \rightarrow 6} \frac{4(x - 6)(x + 6)}{x - 6} \\ &= \lim_{x \rightarrow 6} 4(x + 6) \\ &= 4(12) = 48 \end{aligned}$$

$$(b) \lim_{x \rightarrow 6} \frac{x - 6}{\sqrt{2x - 8} - 2}$$

Explanation.

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 6} \frac{x - 6}{\sqrt{2x - 8} - 2} &= \lim_{x \rightarrow 6} \frac{x - 6}{\sqrt{2x - 8} - 2} \cdot \frac{\sqrt{2x - 8} + 2}{\sqrt{2x - 8} + 2} \\ &= \lim_{x \rightarrow 6} \frac{(x - 6)(\sqrt{2x - 8} + 2)}{2x - 8 - 4} \\ &= \lim_{x \rightarrow 6} \frac{(x - 6)(\sqrt{2x - 8} + 2)}{2(x - 6)} \\ &= \lim_{x \rightarrow 6} \frac{\sqrt{2x - 8} + 2}{2} \\ &= \frac{\sqrt{12 - 8} + 2}{2} \\ &= \frac{4}{2} = 2 \end{aligned}$$

$$(c) \lim_{x \rightarrow 2} \frac{(3x - 2)^2 - 16}{x - 2}$$

Explanation.

$$\begin{aligned}
 & \text{Form: } \frac{0}{0} \\
 \lim_{x \rightarrow 2} \frac{(3x-2)^2 - 16}{x-2} &= \lim_{x \rightarrow 2} \frac{((3x-2)-4)((3x-2)+4)}{x-2} \\
 &= \lim_{x \rightarrow 2} \frac{(3x-6)(3x+2)}{x-2} \\
 &= \lim_{x \rightarrow 2} \frac{3(x-2)(3x+2)}{x-2} \\
 &= \lim_{x \rightarrow 2} 3(3x+2) \\
 &= 3(6+2) = 24
 \end{aligned}$$

$$(d) \lim_{x \rightarrow 6} \frac{\frac{x}{x-2} - \frac{3}{2}}{x-6}$$

Explanation.

Form: $\frac{0}{0}$

$$\begin{aligned}
 \lim_{x \rightarrow 6} \frac{\frac{x}{x-2} - \frac{3}{2}}{x-6} &= \lim_{x \rightarrow 6} \frac{\frac{2x}{2(x-2)} - \frac{3(x-2)}{2(x-2)}}{x-6} \\
 &= \lim_{x \rightarrow 6} \frac{\frac{2x-3x+6}{2(x-2)}}{x-6} \\
 &= \lim_{x \rightarrow 6} \frac{\frac{-x+6}{2(x-2)}}{x-6} \\
 &= \lim_{x \rightarrow 6} \frac{-(x-6)}{2(x-2)(x-6)} \\
 &= \lim_{x \rightarrow 6} \frac{-1}{2(x-2)} \\
 &= \frac{-1}{2(6-2)} = \frac{-1}{2(4)} = \frac{-1}{8}
 \end{aligned}$$

$$(e) \lim_{x \rightarrow 1} \frac{\sqrt{5x-2} - \sqrt{3}}{x-1}$$

Explanation.

$$\begin{aligned}
 & \text{Form: } \frac{0}{0} \\
 \lim_{x \rightarrow 1} \frac{\sqrt{5x-2} - \sqrt{3}}{x-1} &= \lim_{x \rightarrow 1} \frac{\sqrt{5x-2} - \sqrt{3}}{x-1} \cdot \frac{\sqrt{5x-2} + \sqrt{3}}{\sqrt{5x-2} + \sqrt{3}} \\
 &= \lim_{x \rightarrow 1} \frac{(5x-2) - 3}{(x-1)(\sqrt{5x-2} + \sqrt{3})} \\
 &= \lim_{x \rightarrow 1} \frac{5(x-1)}{(x-1)(\sqrt{5x-2} + \sqrt{3})} \\
 &= \lim_{x \rightarrow 1} \frac{5}{\sqrt{5x-2} + \sqrt{3}} \\
 &= \frac{5}{\sqrt{5(1)-2} + \sqrt{3}} \\
 &= \frac{5}{2\sqrt{3}}
 \end{aligned}$$

Problem 3

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Problem 3 Evaluate the following limits.

(a) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos(x)}{x}$

Explanation. Apply the Quotient Law and use continuity of the numerator and denominator.

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos(x)}{x} = \frac{0}{\frac{\pi}{2}} = 0$$

(b) $\lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 4}$

Explanation. Apply the Quotient Law and use continuity.

$$\lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 4} = \frac{\sqrt{25 - (3)^2} - 4}{3 - 4} = \frac{4 - 4}{-1} = 0$$

(c) $\lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3}$

Explanation. Since $\lim_{x \rightarrow 3} (\sqrt{25 - x^2} - 4) = 0$ and $\lim_{x \rightarrow 3} x - 3 = 0$, this limit has form $\frac{0}{0}$.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3} &= \lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3} \cdot \frac{\sqrt{25 - x^2} + 4}{\sqrt{25 - x^2} + 4} = \\ &= \lim_{x \rightarrow 3} \frac{25 - x^2 - 16}{(x - 3)(\sqrt{25 - x^2} + 4)} = \lim_{x \rightarrow 3} \frac{9 - x^2}{(x - 3)(\sqrt{25 - x^2} + 4)} = \lim_{x \rightarrow 3} \frac{(3 + x)(3 - x)}{(x - 3)(\sqrt{25 - x^2} + 4)} = \\ &= \lim_{x \rightarrow 3} \frac{-(3 + x)}{\sqrt{25 - x^2} + 4} = \frac{-6}{\sqrt{25 - 9} + 4} = \frac{-6}{\sqrt{16} + 4} = \frac{-6}{8} = \frac{-3}{4} \end{aligned}$$

Problem 4

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Problem 4

Problem 4 Two polynomials, h and g , are given

$$h(x) = \frac{x^2 - 4}{4}$$

$$g(x) = x - 2$$

State the form of the limit, evaluate the limit, or state that it does not exist. Justify your answer.

(a) $\lim_{x \rightarrow 2} \frac{h(x)}{g(x)}$

Explanation. $\lim_{x \rightarrow 2} \frac{h(x)}{g(x)} = \lim_{x \rightarrow 2} \frac{\frac{x^2 - 4}{4}}{x - 2}$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{\frac{x^2 - 4}{4}}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{4(x - 2)} \\ &= \lim_{x \rightarrow 2} \frac{x + 2}{4} \\ &= \frac{4}{4} \\ &= 1 \end{aligned}$$

(b) $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3}$

Explanation. $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3} = \lim_{x \rightarrow 3} \frac{x - 2 - 1}{x - 3}$

Form: $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3} &= \lim_{x \rightarrow 3} \frac{x - 2 - 1}{x - 3} \\ &= \lim_{x \rightarrow 3} 1 \\ &= 1 \end{aligned}$$

(c) $\lim_{x \rightarrow 4} \frac{h(x) - h(4)}{x - 4}$

Explanation. $\lim_{x \rightarrow 4} \frac{h(x) - h(4)}{x - 4} = \lim_{x \rightarrow 4} \frac{\frac{x^2 - 4}{4} - 3}{x - 4}$

Form: $\frac{0}{0}$

Problem 4

$$\begin{aligned}\lim_{x \rightarrow 4} \frac{h(x) - h(4)}{x - 4} &= \lim_{x \rightarrow 4} \frac{\frac{x^2 - 4}{4} - 3}{x - 4} \\&= \lim_{x \rightarrow 4} \frac{x^2 - 4 - 12}{4(x - 4)} \\&= \lim_{x \rightarrow 4} \frac{x^2 - 16}{4(x - 4)} \\&= \lim_{x \rightarrow 4} \frac{(x - 4)(x + 4)}{4(x - 4)} \\&= \lim_{x \rightarrow 4} \frac{x + 4}{4} \\&= 2\end{aligned}$$

Problem 5

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Problem 5 Evaluate each of the following limits. Possible answers include a number, $+\infty$, $-\infty$ and “Does Not Exist” (DNE). Make sure to state the form of the limit. Justify your answer.

(a) $\lim_{x \rightarrow 3^-} \frac{x^2 - 3}{x^2 - x - 6}$

Explanation. $\lim_{x \rightarrow 3^-} \frac{x^2 - 3}{x^2 - x - 6} = \lim_{x \rightarrow 3^-} \frac{x^2 - 3}{(x - 3)(x + 2)}$ is of the form $\frac{\#}{0}$.

Since $\lim_{x \rightarrow 3^-} (x^2 - 3) = 6$, and $\lim_{x \rightarrow 3^-} (x^2 - x - 6) = \lim_{x \rightarrow 3^-} (x - 3)(x + 2) = 0$, and since

$$x^2 - x - 6 = (x - 3)(x + 2) < 0, \text{ if } x < 3 \text{ and } x \text{ close to } 3.$$

$$\lim_{x \rightarrow 3^-} \frac{x^2 - 3}{x^2 - x - 6} = -\infty$$

(b) $\lim_{x \rightarrow 5^+} \frac{x^2 + 6}{x^2 - 3x - 10}$

Explanation. The limit is of the form $\frac{\#}{0}$.

$$\lim_{x \rightarrow 5^+} \frac{x^2 + 6}{x^2 - 3x - 10} = \lim_{x \rightarrow 5^+} \frac{x^2 + 6}{(x - 5)(x + 2)} = \infty, \text{ because}$$

$\lim_{x \rightarrow 5^+} (x^2 + 6) = 31$ and $x^2 - 3x - 10 = (x - 5)(x + 2) > 0$, for $x > 5$ and x close to 5.

(c) $\lim_{x \rightarrow 1} \frac{4 - x}{x^2 - 2x + 1}$

Explanation. The limit is of the form $\frac{\#}{0}$.

Checking left and right sided limits we see:

$$\lim_{x \rightarrow 1^+} \frac{4 - x}{x^2 - 2x + 1} = \lim_{x \rightarrow 1^+} \frac{4 - x}{(x - 1)^2} = \infty$$

and

$$\lim_{x \rightarrow 1^-} \frac{4 - x}{x^2 - 2x + 1} = \lim_{x \rightarrow 1^-} \frac{4 - x}{(x - 1)^2} = \infty$$

$$\text{Since } \lim_{x \rightarrow 1^-} \frac{4 - x}{x^2 - 2x + 1} = \lim_{x \rightarrow 1^+} \frac{4 - x}{x^2 - 2x + 1} \implies \lim_{x \rightarrow 1} \frac{4 - x}{x^2 - 2x + 1} = \infty$$

(d) $\lim_{x \rightarrow 2} \frac{-e^x}{(2-x)^3}$

Explanation. The limit is of the form $\frac{\#}{0}$.

Checking right limit we have:

$$\lim_{x \rightarrow 2^+} \frac{-e^x}{(2-x)^3} = \infty$$

and checking the left limit we have:

$$\lim_{x \rightarrow 2^-} \frac{-e^x}{(2-x)^3} = -\infty$$

Since $\lim_{x \rightarrow 2^+} \frac{-e^x}{(2-x)^3} \neq \lim_{x \rightarrow 2^-} \frac{-e^x}{(2-x)^3} \implies \lim_{x \rightarrow 2} \frac{-e^x}{(2-x)^3}$ Does not exist

(e) $\lim_{x \rightarrow 1} \frac{\sin x}{\sqrt{2-x^2}-1}$

Explanation. The limit is of the form $\frac{\#}{0}$.

Checking right limit we have:

$$\lim_{x \rightarrow 1^+} \frac{\sin x}{\sqrt{2-x^2}-1} = -\infty,$$

$\lim_{x \rightarrow 1^+} \sin x = \sin 1 > 0$, since $1 < \frac{\pi}{2}$. Explanation: Note that for x near 1 and such that $x > 1$, we have that

$x^2 > 1$, and by multiplying by (-1) , we get

$-x^2 < -1$, and by adding 2 on both sides, we get

$2 - x^2 < 2 - 1$, and by taking the square root, we get

$\sqrt{2-x^2} < \sqrt{1}$, or

$\sqrt{2-x^2} < 1$, and by subtracting 1 from both sides, we get

$\sqrt{2-x^2}-1 < 0$.

Therefore the numerator is positive, and denominator is **negative** and goes to 0.

Problem 5

Checking the left limit we have: $\lim_{x \rightarrow 1^-} \frac{\sin x}{\sqrt{2-x^2}-1} = +\infty$

Explanation: Note that and for x near 1 such that $x < 1$, we have that

$x^2 < 1$, and by multiplying by (-1) , we get

$-x^2 > -1$, and by adding 2 on both sides, we get

$2 - x^2 > 2 - 1$, and by taking the square root, we get

$\sqrt{2 - x^2} > \sqrt{1}$, or

$\sqrt{2 - x^2} > 1$, and by subtracting 1 from both sides, we get

$\sqrt{2 - x^2} - 1 > 0$.

Therefore the numerator is positive, and denominator is **positive** and goes to 0.

Since $\lim_{x \rightarrow 1^+} \frac{\sin x}{\sqrt{2-x^2}-1} \neq \lim_{x \rightarrow 1^-} \frac{\sin x}{\sqrt{2-x^2}-1} \implies \lim_{x \rightarrow 1} \frac{\sin x}{\sqrt{2-x^2}-1}$ does not exist.

Problem 6

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Problem 6 A piecewise defined function f is given by

$$f(x) = \begin{cases} \frac{2x-3}{x-2} & \text{if } x < 2 \\ \frac{x^2-5x+6}{x^2-4} & \text{if } x > 2 \end{cases}$$

Determine the form of the limit, then find the limit.

(a) $\lim_{x \rightarrow 2^+} f(x)$

Explanation. Since $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{x^2-5x+6}{x^2-4}$, the form of the limit is $\frac{0}{0}$.

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{x^2-5x+6}{x^2-4} = \lim_{x \rightarrow 2^+} \frac{(x-3)(x-2)}{(x-2)(x+2)} = \lim_{x \rightarrow 2^+} \frac{x-3}{x+2} = -\frac{1}{4}$$

(b) $\lim_{x \rightarrow 2^-} f(x)$

Explanation. Since $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{2x-3}{x-2}$, the form of the limit is $\frac{\#}{0}$.

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{2x-3}{x-2} = -\infty,$$

since the numerator positive, the denominator negative and goes to 0.

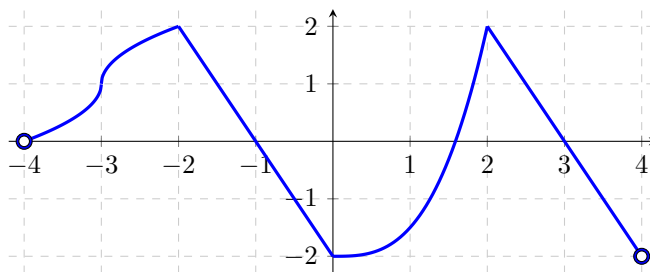
(c) $\lim_{x \rightarrow 2} f(x)$

Explanation. The limit does not exist, because $\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$.

Problem 7

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Problem 7 The graph of a function $g(x)$ with domain $(-4, 4)$ is given in the figure. This portions of this graph in the intervals $(-2, 0)$ and $(2, 4)$ are straight lines. The portion on the interval $(0, 2)$ is not parabolic.



For each of the limits below, give the form of the limit, then evaluate the limit.

(a) $\lim_{x \rightarrow -3} \frac{g(x) + \sin\left(\frac{\pi}{4}x\right)}{x\sqrt{x+4}}.$

Explanation. The numerator and denominator are each continuous at $x = -3$, and at $x = -3$ the denominator is nonzero. The entire fraction is continuous at $x = -3$. To find the limit, we can plug in the value. (We would call this form either “ $\frac{\#}{\#}$ ” or “Continuous”).

$$\begin{aligned} \lim_{x \rightarrow -3} \frac{g(x) + \sin\left(\frac{\pi}{4}x\right)}{x\sqrt{x+4}} &= \frac{g(-3) + \sin\left(\frac{-3\pi}{4}\right)}{-3\sqrt{-3+4}} \\ &= \frac{1 - \frac{1}{\sqrt{2}}}{-3}. \end{aligned}$$

(b) $\lim_{x \rightarrow 0^+} \frac{e^x}{|g(x) + 2|}.$

Explanation. $\lim_{x \rightarrow 0^+} e^x = e^0 = 1$ and $\lim_{x \rightarrow 0^+} |g(x) + 2| = 0$, so this limit has form $\frac{\#}{0}$. One-sided limits with form $\frac{\#}{0}$ are either $+\infty$ or $-\infty$, so we need to check the sign of the fraction.

We know the range of the famous function e^x is $(0, \infty)$, so the numerator here is always positive. The denominator is an absolute value, so it is not negative either. As long as x is near, but not equal, to 0, $g(x) + 2$ will be close, but not equal, to 0, so $|g(x) + 2|$ will be positive.

$$\lim_{x \rightarrow 0^+} \frac{e^x}{|g(x) + 2|} = \infty.$$

$$(c) \lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x + g(x) - 3}$$

Explanation. $\lim_{x \rightarrow 3} (x^2 + 2x - 15) = 0$ and $\lim_{x \rightarrow 3} (x + g(x) - 3) = 0$, so this limit has form $\frac{0}{0}$. In order to use algebra to simplify this fraction, we need a formula for $g(x)$ for x near 3. From the graph we see that in the interval $(2, 4)$, the graph of g is a straight line with slope -2 . The equation of that line is $y = -2x + 6$, so for x close to 3, we know $g(x) = -2x + 6$.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x + g(x) - 3} &= \lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x + (-2x + 6) - 3} \\ &= \lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{-x + 3} \\ &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 5)}{-(x - 3)} \\ &= \lim_{x \rightarrow 3} -(x + 5) \\ &= -8. \end{aligned}$$